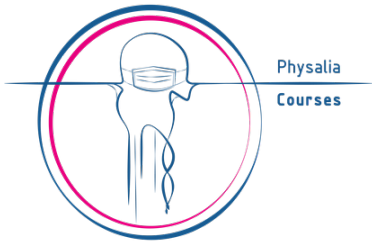
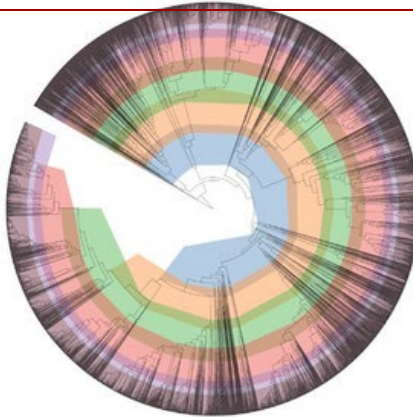


ENVIRONMENTAL METAGENOMICS

Physalia course, online, 11-15 November 2024

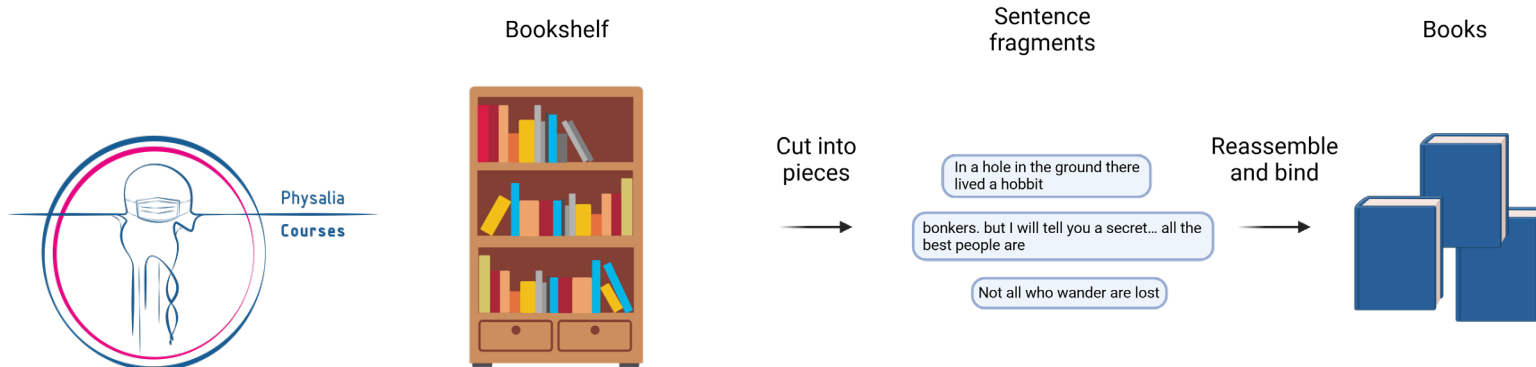
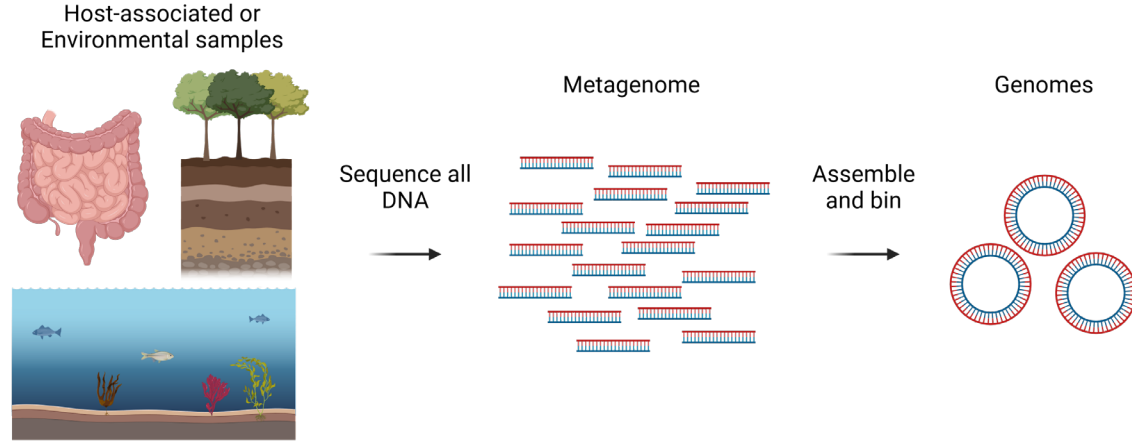
MAG binning

Nikolay Oskolkov, Lund University, NBIS SciLifeLab
Samuel Aroney, Queensland University of Technology



NB: original course material courtesy:
Dr. Antti Karkman, University of Helsinki
Dr. Igor Pessi, Finnish Environment Institute (SYKE)
As. Prof. Luis Pedro Coelho

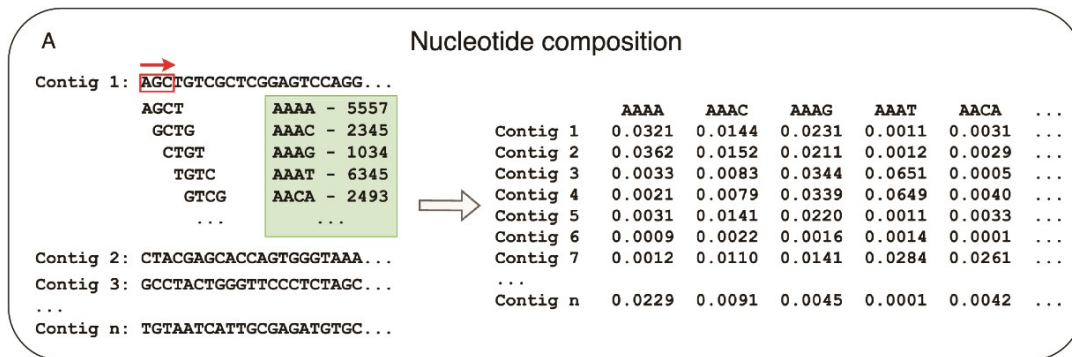
Binning: collating contigs into collections



Standard binning strategies

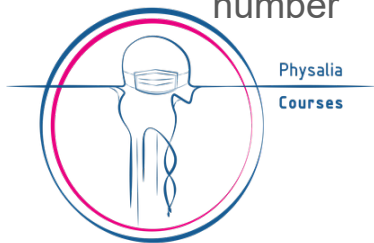
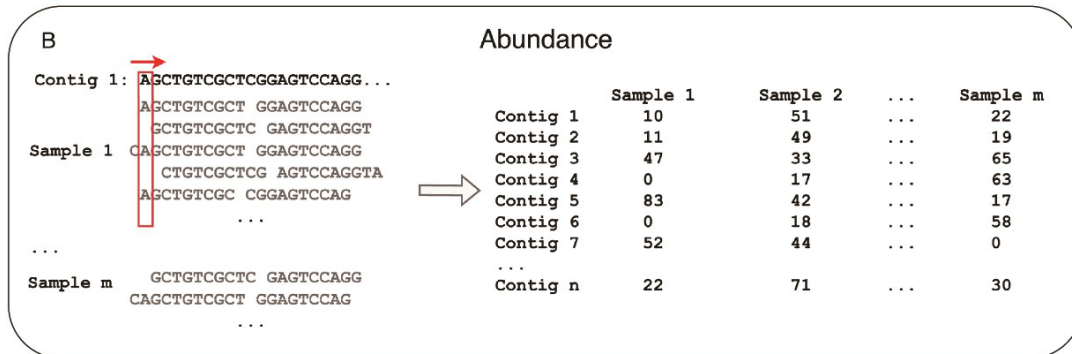
- Nucleotide composition

- Microbes tend to have similar %GC and codon choice across genome



- Abundance

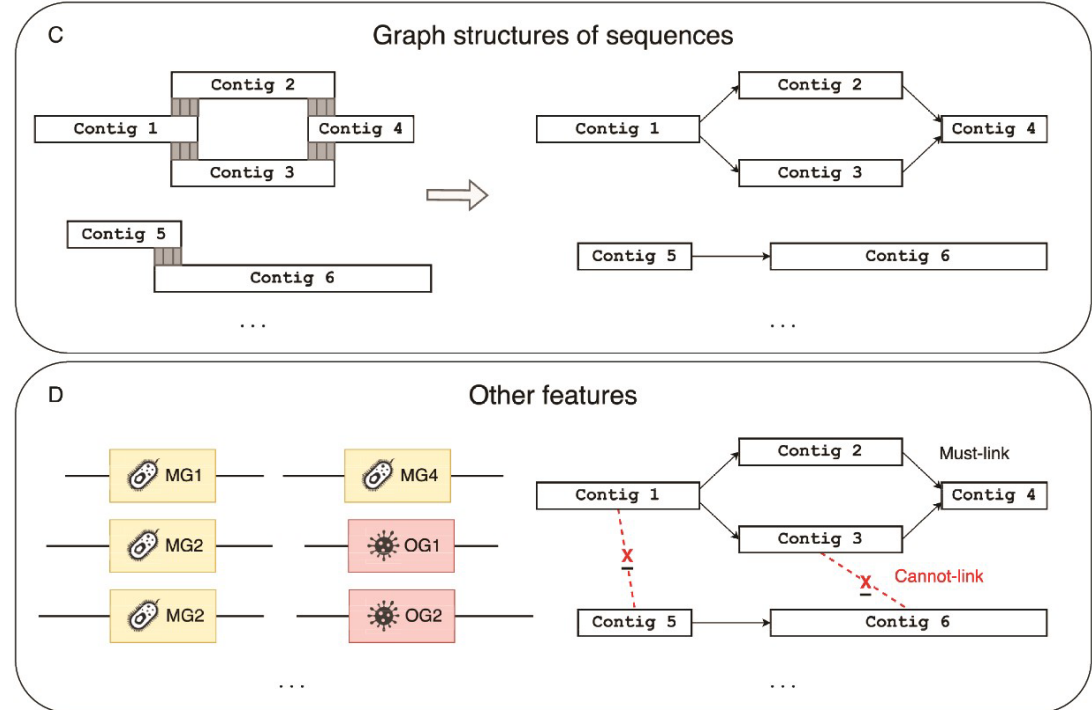
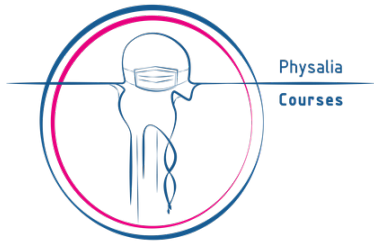
- The whole genome should be present at similar copy number



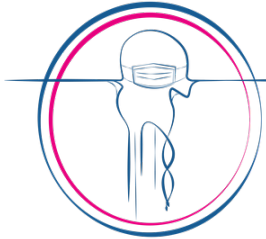
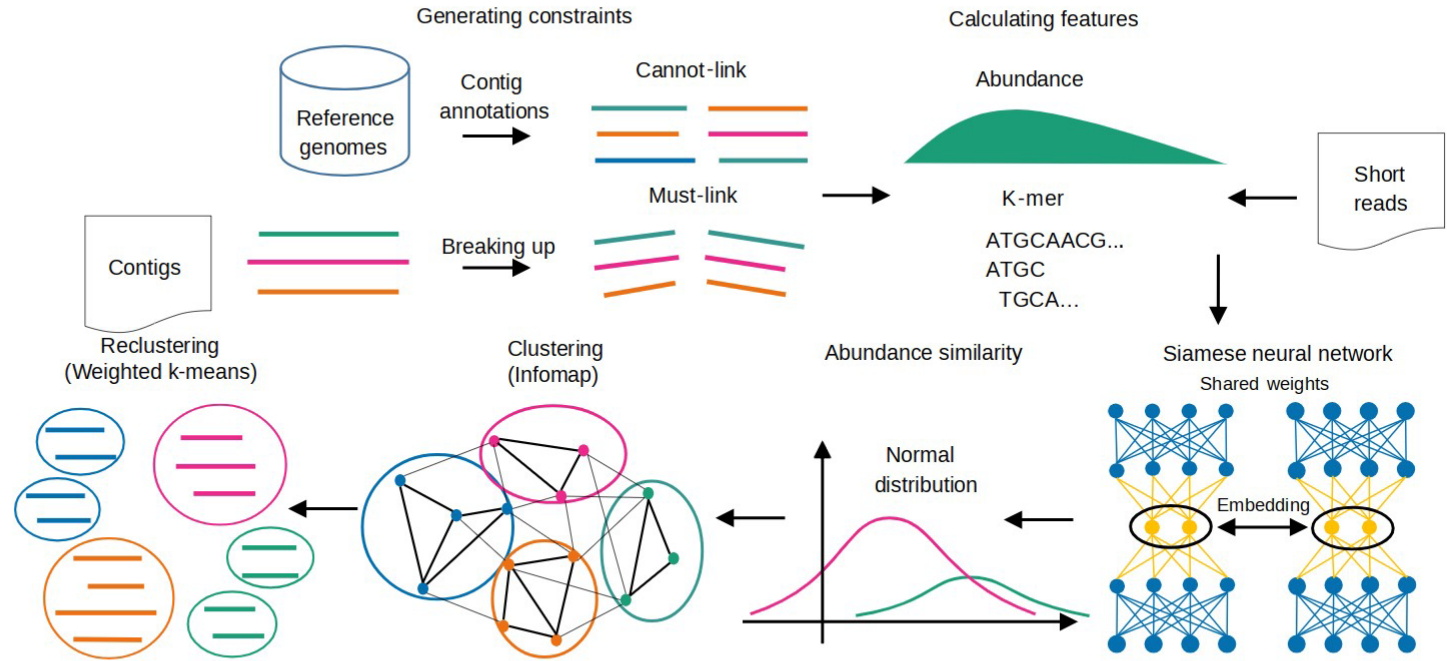
Physalia
Courses

More binning strategies

- Graph structures
 - Assembly graph structures can be reused
- Other features
 - Methylation patterns
 - Single-copy marker genes
 - Machine learning inputs



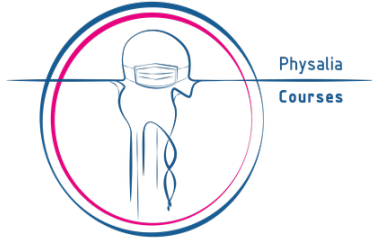
Semibin



Learning a new model is resource intensive

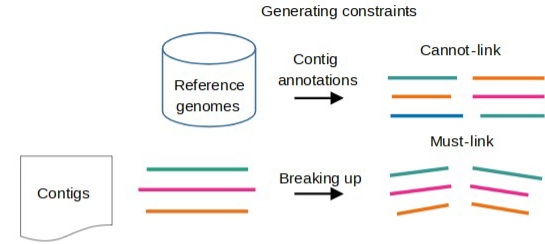
- SemiBin uses a learned model
- Generating the data for training is expensive (mmseqs2 is slow)
- Learning is slow (a GPU helps, but few people have a GPU cluster)

		Single-sample		
		Human gut	Dog gut	Ocean
Computing features	Time(min)	5	9	12
	Memory(MB)	923	736	1,675
Generating cannot-link	Time(min)	88	81	114
	Memory(MB)	39,070	37,904	46,091
Training(CPU)	Time(min)	181	209	222
	Memory(MB)	2,497	2,373	3,211
Training(GPU)	Time(min)	34	36	45
	Memory(MB)	4,487	4,355	5,222
Binning	Time(min)	2	2	3
	Memory(MB)	4,501	3,641	7,622



From Semi- to self-supervised learning

- In SemiBin1, must-links were self-supervised
- In SemiBin1, cannot-links were from a reference taxonomy



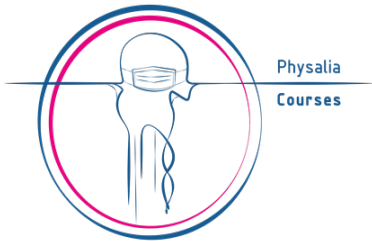
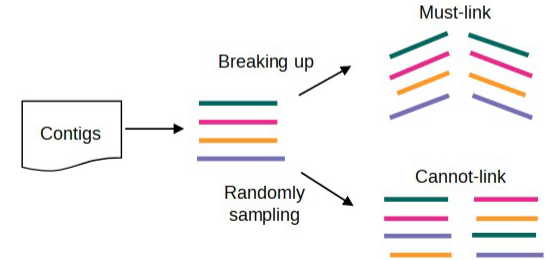
Use random sampling to generate cannot-links

Random sampling advantages

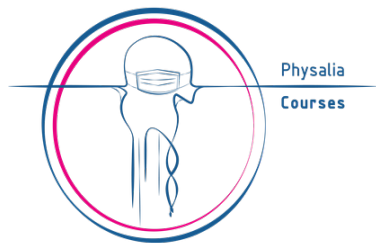
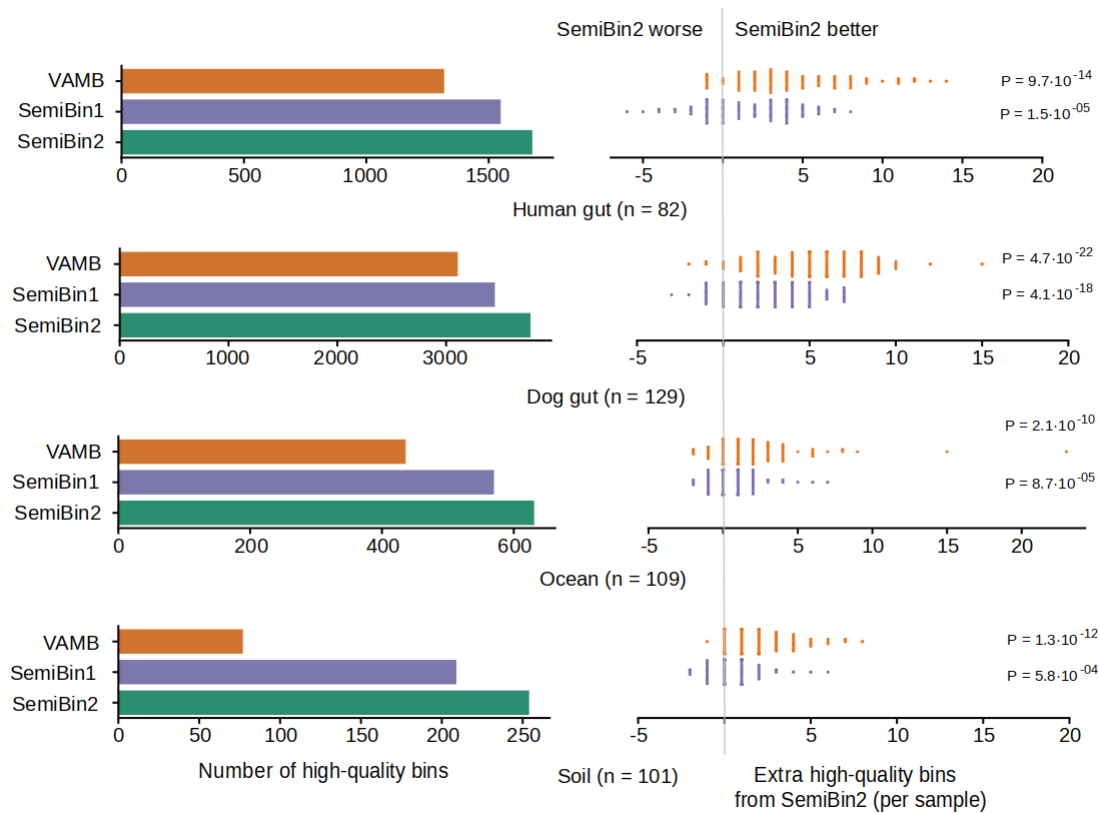
1. Random sampling is faster than MMSeqs2
2. It has no database biases

Random sampling disadvantages

1. Random sampling introduces errors (naturally)

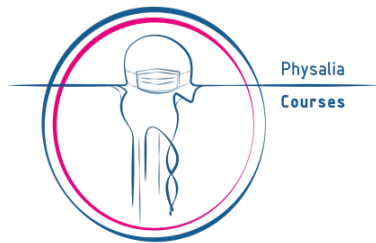
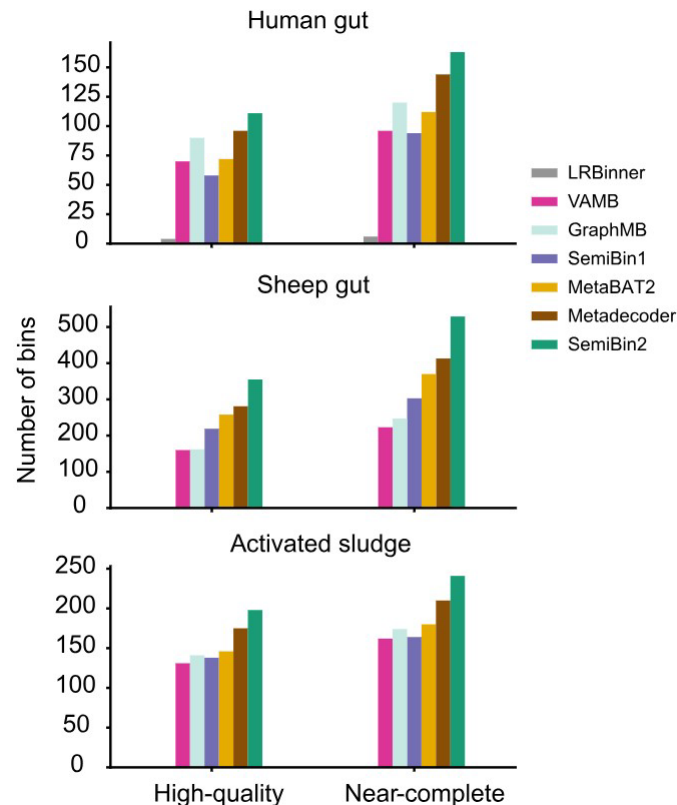


Semibin vs Semibin2



SemiBin2 surpasses alternatives for long-read binning

- Three datasets: human gut, sheep gut, and activated sludge
- Some of the alternatives (like SemiBin1) are not designed for long-reads
- We used [CheckM2](#) & [GUNC](#) to evaluate the results
- SemiBin2 is also best using [checkM1](#), but the two tools share some of the same marker genes (so evaluation is partly circular)



All the binners

Binning metagenomic contigs by coverage and composition

Johannes Alneberg^{1,8}, Brynjar Smári Bjarnason^{1,8}, Ino de Bruijn^{1,2}, Melanie Schirmer³, Joshua Quick^{4,5}, Umer Z Ijaz³, Leo Lahti^{6,7}, Nicholas J Loman⁴, Anders F Andersson^{1,9} & Christopher Quince^{3,9}

MetaBAT, an efficient tool for accurately reconstructing single genomes from

MetaBAT 2: an adaptive binning algorithm for robust and efficient genome reconstruction from metagenome assemblies

Dongwan D. Kang¹, Feng Li², Edward Kirton¹, Ashleigh Thomas¹, Rob Egan¹, Hong An² and Zhong Wang^{1,3,4}

Effective binning of metagenomic contigs using contrastive multi-view representation learning

Received: 28 June 2023

Ziye Wang¹, Ronghui You¹, Haitao Han¹, Wei Liu¹, Fengzhu Sun² &

Shanfeng Zhu^{1,3,4,5,6}

Accepted: 7 December 2023

A de novo metagenomic binning method for large datasets across different environments

Shaojun Pan^{1,2}, Chengkai Zhu^{1,2,3}, Xing-Ming Zhao^{1,2,4,5} & Luis Pedro Coelho^{1,2}

MaxBin 2.0: an automated binning algorithm to recover genomes from multiple metagenomic datasets

Yu-Wei Wu^{1,2,*}, Blake A

Improved metagenome binning and assembly using deep variational autoencoders

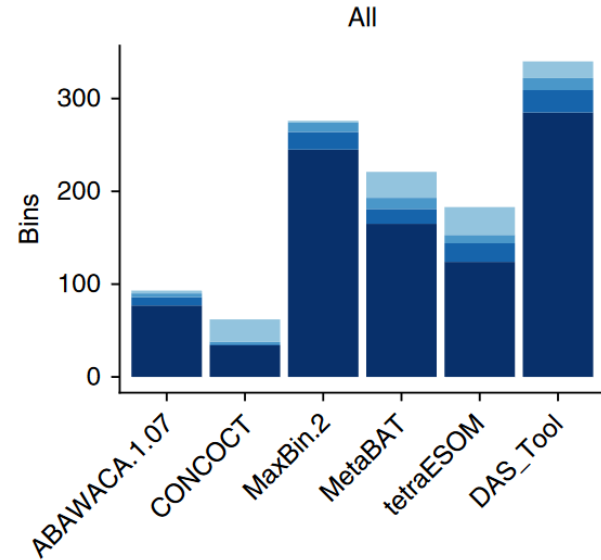
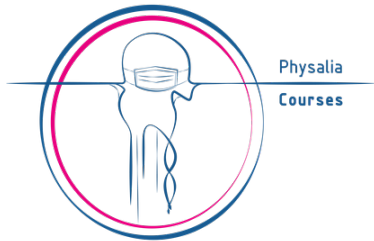
Jakob Nybo Nissen^{1,2}, Joachim Johansen², Rosa Lundbye Allesøe², Casper Kaae Sønderby³, Jose Juan Almagro Armenteros¹, Christopher Heje Grønbech^{3,4}, Lars Juhl Jensen², Henrik Bjørn Nielsen⁵, Thomas Nordahl Petersen⁵, Ole Winther^{3,4,7} and Simon Rasmussen²

SemiBin2: self-supervised contrastive learning leads to better MAGs for short- and long-read sequencing

Shaojun Pan^{1,2}, Xing-Ming Zhao^{1,2,3,4,*}, Luis Pedro Coelho^{1,2,*}

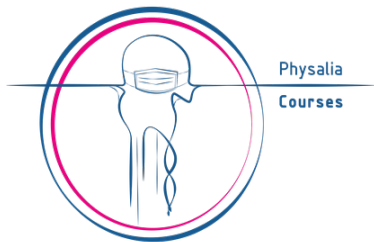
Ensemble binning

- Combining bins from multiple binner
- DAS Tool
- MetaWRAP
- MetaBinner
- BASALT



Aviary: Our Assembly + Binning pipeline

- Snakemake pipeline (automatically submit jobs to a cluster)
- Assembly: metaSPAdes or MegaHit
 - Particularly good at hybrid short+long read
- Binning
 - MetaBAT1/2, VAMB, SemiBin1/2, CONCOCT, MaxBin2, Rosella
 - DAS Tool



All the binner - comparison

semibin = Semibin2

